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MILITARY SPECIFICATION

SPECTROMETER, FLUID ANALYSIS A/E35U-3

This specification is mandatory for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fluid analysis spectrometer, designated A/E35U-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CC-M-635	Motors, Fractional Horsepower (Alternating-Current)
CC-M-641	Motors, Alternating Current (Integral Horsepower, 200 HP and Smaller)

Military

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173	Varnish, Moisture- and Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-5015	Connectors, Electric, AN Type
MIL-T-5021	Test, Aircraft and Missile Welding Operators' Qualification
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for

FSC 6650

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MIL-A-8421	Air Transportability Requirements, General Specification for
MIL-S-8516	Sealing Compound, Synthetic Rubber, Electric Connectors and Electric Systems, Accelerator Required
MIL-T-21200	Test Equipment for Use with Electronic and Electrical Equipment, General Specification for
MIL-I-26860	Indicator, Humidity, Plug, Color Change
MIL-S-38130	Safety Engineering of Systems and Associated Subsystems, and Equipment, General Requirements for

STANDARDS

Federal

595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	Maintainability Demonstration
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-838	Lubrication of Military Equipment
MIL-STD-889	Dissimilar Metals
MIL-STD-1247	Marking, Functions and Hazard Designations of Hose, Pipe, and Tube Lines for Aircraft, Missile, and Space Systems
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute

C1-1968	National Electrical Code
S1.4-1961	General Purpose Sound Level Meters
S1.11-1966	Octave, Half-Octave, and Third-Octave Band Filter Sets

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

National Bureau of Standards

Handbook H28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

Naval Air Rework Facility Drawing

B-34994	Sampling Kit, Spectrometric Oil Analysis

(Application for copies should be addressed to the Naval Air Rework Facility, Pensacola, Florida.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials used in the construction of the spectrometer shall be as specified herein. Materials shall not be hammered, filled, nor otherwise treated to conceal defects. Materials not definitely specified shall be of uniform quality and condition and free of seams, cracks, or other defects which may adversely affect the appearance, strength, endurance, and wear resistance of the spectrometer.

3.3.1 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically

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sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.1.1 Electrical components. Electrical components shall be treated for fungus resistance in accordance with MIL-T-152. Varnish shall conform to MIL-V-173.

3.3.2 Protective treatment. When materials are used in the construction of the spectrometer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic and environmental conditions shall not be used.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, oil, salt fog, and atmospheric conditions likely to be met in storage and normal service.

3.3.3.1 Dissimilar metals. Unless protected from electrolytic action of corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.4 Design and construction. The spectrometer, exclusive of associated printout equipment, shall be a single self-contained package that requires only the application of necessary utilities to provide the required performance. The spectrometer shall be so designed and constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation, and service.

3.4.1 Design principles. The principles of human engineering as specified in MIL-STD-1472 shall be applied in the design of the spectrometer to provide the most efficient environment practical for the performance of the mission whether installed aboard ship, in a shop, or van. Controls shall require as few movements as possible and shall be consistent in direction of motion for all equipment which the operator uses. Displays shall be simple in design and present the information in the most readable form. Every effort shall be made to minimize undesirable side effects on personnel produced by the equipment or the material under test, such as vibration, noise, noxious gases, or radiation.

3.4.1.1 Modular construction. Solid-state modular construction with plug-in features shall be used to the extent necessary to enable maintenance at as low a level as possible to quickly correct faulty operation by replacement of an entire system or subsystem.

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3.4.1.2 Self-test provisions. The design of the spectrometer shall incorporate integral, complete, rapid, diagnostic self-test provisions to enable the operator to observe whether the spectrometer is operating within tolerances down to the modular unit level.

3.4.1.3 Operator training considerations. The spectrometer shall be so designed that operation, calibration, and routine maintenance is within the capability of the operator after a short training period, not exceeding 2 weeks, using only appropriate manuals supplied with the spectrometer. The spectrometer shall be so designed that it can be operated by one person. Training aids and devices shall not be required.

3.4.2 Reliability. The spectrometer shall be designed to operate at a rate of 40-analyses per hour during missions of 11.5 hours with 0.5 hour turnaround time. Mean-time-between-failures (MTBF) under these conditions will be the parameter used to express reliability. Failure rate, the inverse of MTBF, will be assumed constant (see 6.3.1 and 6.3.2).

3.4.2.1 Design reliability. The spectrometer design, excluding associated printout equipment, shall have a predicted MTBF of 450 hours, or more, based on the failure rate prediction required by MIL-STD-781. Documented component failure rate data or estimation rationale used as a basis for the prediction shall be available.

3.4.2.2 Specified reliability. The spectrometer, excluding associated printout equipment, shall have a demonstrable specified MTBF ($\emptyset_o$ as defined in MIL-STD-781) of at least 225 hours.

3.4.2.3 Achieved availability. The achieved availability shall be 94 percent, exclusive of movement time (see 6.3.3).

3.4.3 Maintainability. The spectrometer shall be designed and constructed to conform to the design guidelines of MIL-STD-470, paragraph entitled Establish maintainability design criteria. Except rather than being merely considered, the guidelines shall apply wherever appropriate to the subject class or type of equipment and wherever they do not conflict with other requirements specified herein. All maintenance, servicing, and adjustments which may be economically practicable at lower than depot level shall be performed using only common handtools and equipment commonly expected in shops doing maintenance on similar equipment.

3.4.3.1 Maintenance downtimes (see 6.3.5). The spectrometer shall be designed and constructed to permit maintenance by military maintenance technicians and operators in not more than the maintenance times specified below. The requirements shall apply individually to both on-equipment and off-equipment maintenance to be done below depot or equivalent level. The level of component repair and

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replacement shall be defined, based on contractor recommendations or, if applicable, the approved maintenance and provisioning plans. Not more than one technician shall be required for each task.

3.4.3.1.1 Corrective maintenance downtime. The mean corrective maintenance downtime shall not exceed 1 hour with a related maximum corrective maintenance downtime not exceeding 3 hours.

3.4.3.1.2 Total preventive and corrective maintenance. The accumulated preventive and corrective maintenance manhours, as defined by MIL-STD-721 shall not exceed an average of 0.06 of the accumulated operating hours during the design useful operating life of the equipment (eg, not more than 180 man-hours in 3,000 operating hours). This total preventive and corrective maintenance time shall include servicing tasks other than normal and planned operator tasks requiring less than 5 minutes, such as cleaning light transmission windows or replacing electrodes.

3.4.3.1.3 Preventive maintenance. The preventive maintenance schedule and estimated frequencies shall be established based on the recommended or, if applicable, approved maintenance manuals, on component vendor recommendations, and on other data and engineering analyses. Preventive maintenance task times shall be apportioned to comply with the above quantitative requirement and to provide a preventive maintenance requirement for demonstration.

3.4.4 Capability. The spectrometer shall be designed for analyzing both petroleum and synthetic base samples, and low flashpoint (200°F minimum) materials such as hydraulic fluids. The spectrometer shall be capable of analyzing any one sample to determine the concentration value of each of the 20 elements in parts per million (ppm) as related to the standards provided for calibration. The order of element instrument readout of the 20 elements is as follows:

Iron	(Fe)	Silicon	(Si)
Silver	(Ag)	Tin	(Sn)
Aluminum	(Al)	Titanium	(Ti)
Beryllium	(Be)	Boron	(B)
Chromium	(Cr)	Barim	(Ba)
Copper	(Cu)	Cadmium	(Cd)
Magnesium	(Mg)	Manganese	(Mn)
Sodium	(Na)	Molybdenum	(Mo)
Nickel	(Ni)	Vanadium	(V)
Lead	(Pb)	Zinc	(Zn)

Note. The phrase "as related to the standards provided for calibration" is construed to mean that when any calibration standard (eg, 100 ppm by weight) is analyzed, the concentration displayed either on the indicator (see 3.9.2) or printed out by the associated printout equipment shall be the concentration value of the standard (eg, 100) within the allowable tolerance of 3.9.1.1.

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3.4.4.1 The spectrometer shall be capable of analyzing a sample for all elements required without human preparation of the sample prior to analysis. If a container is required, the spectrometer shall be capable of analyzing samples in the cap of a standard sample container (1-1/16 inch OD by 1 inch high as shown on Drawing B-34994, note 2), as received in the laboratory. Other containers may be utilized if required for special applications.

3.4.4.2 Operating capacity. The spectrometer shall be capable of continuously analyzing a minimum of 40 samples per hour. The time required for the analysis and automatic readout of one sample for 20 elements shall not exceed 1.5 minutes.

3.4.4.3 Sample size. The spectrometer shall be designed to require not more than 5 milliliters for complete sample analysis.

3.4.5 Safety. The spectrometer shall be so designed and constructed that all operating and maintenance conditions will be class I (safe) or class II (marginal) in accordance with MIL-S-38130. The paragraphs of MIL-S-38130 entitled System safety criteria (individual paragraph only) and System safety sequence (including subparagraphs) shall apply where hazards are inherent in the function performed and shall apply in design features. Operating personnel shall be protected from dangerous voltages and radiation.

3.4.5.1 Oil sample burning. Operator protection shall be provided during oil sample burning.

3.4.5.2 Noxious fumes. The spectrometer shall be provided with means for ingesting fresh air from, and expelling to the outside any noxious fumes which may result from the analysis. The exhaust system shall be capable of expelling the fumes through a minimum of 50 feet of ducting with two 90° elbows and without the use of an external exhaust fan. The airflow shall be controlled by a damper and shown by a draft indicator.

3.4.6 Compactness. Compact, lightweight components shall be used throughout the fabrication of the spectrometer.

3.4.7 Lubrication. All points of friction of the spectrometer shall be lubricated in accordance with MIL-STD-838 to insure satisfactory operation.

3.4.7.1 Frequency of lubrication. The spectrometer shall be so designed that it will not require lubrication at intervals of less than 30 days when operated a minimum of 8 hours per day.

3.4.8 Installation. The spectrometer and transit case specified in 3.12 shall be so designed and constructed that the spectrometer can be removed from the transit case and installed for operation within 8 man-hours.

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3.4.9 Air transportability. The spectrometer, installed in the transit case, shall be designed and constructed in accordance with MIL-A-8421.

3.4.10 Plumbing. All lines, valves, and fittings shall be of sufficient size to permit operation under maximum flow conditions. All lines, ducts, and tubing shall be identified in accordance with MIL-STD-1247.

3.4.10.1 Rigid lines. The use of rigid lines shall be held to a minimum. When used, such lines shall be of corrosion-resistant steel or aluminum alloy. Rigid lines, fittings, and components shall withstand 200 percent of the maximum operating pressure without leakage or permanent distortion. Commercial black pipe and fittings shall not be used. Materials (other than thread lubrication and teflon tape) shall not be used for sealing threaded joints.

3.4.10.2 Flexible lines and fittings. Flexible AN or MS hose with AN- or MS-type fittings shall be used. Hose clamps shall not be used for fluid line couplings. All fluid lines, valves, and fittings shall be of a type rated by their manufacturers for the pressure and service to be encountered and shall withstand 200 percent of maximum operating pressure without leakage or permanent distortion.

3.4.11 Welding. Welding shall be accomplished by operators certified in accordance with class A of MIL-T-5021.

3.4.12 Installations. The spectrometer is intended for installation and service at the following locations:

a. On board a ship - The spectrometer in the installed position shall meet the requirements of MIL-S-901, grade B

b. At a fixed land facility

c. In a mobile M119, 6-ton, two-wheel cargo semitrailer van.

3.5 Performance. The spectrometer shall require not more than 1 hour for stabilization after initial power is applied and shall be capable of analyzing samples to determine the contaminant value (in ppm) of the elements specified herein under all operational service conditions specified in 3.5.1.

3.5.1 Operational conditions. The spectrometer shall be capable of operating under the following conditions:

a. Temperatures ranging from 50° to 110°F

b. Low pressure up to 10,000 feet above sea level

c. Vibration in accordance with MIL-STD-167, type 1

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d. Operation in an explosive atmosphere

e. Shipboard roll of 10° and 5° pitch; permanent list of 10°, and 5° trim fore and aft

f. Tilted on a 5° incline.

3.5.2 Nonoperational conditions. The spectrometer shall be capable of operating after being subjected to the following conditions:

a. Storage temperatures ranging from -80° to +160°F

b. Low pressure at 50,000 feet above sea level

c. Vibration as encountered in transportation and handling

d. Shock as encountered in transportation and handling

e. Movement in installed position (semitrailer van) over unimproved roads and off-roads over rolling terrain

f. Installed position (shipboard), roll of 45°, pitch and list of 10°

g. Salt fog atmosphere

h. Dust particles as encountered in dry climates

i. Fungus growth as encountered in tropical climates

j. Relative humidity up to 95 percent, including conditions wherein condensation takes place in the form of water.

3.6 Electrical system. The spectrometer shall operate from 120/208 and 240/416V ac ±20 percent, 50 and 60 Hz ±5 percent, single phase, three-wire power. The electrical system shall accommodate voltage transient conditions of ±20 percent in a period of 1 second.

3.6.1 Control devices

3.6.1.1 Safety switch. A safety device shall be provided that will prevent damage to the spectrometer in the event of an erroneous voltage condition. The electrical system shall be so designed that when the spectrometer is wired for operation from one of the voltage and frequency combinations specified in 3.6, the application of any other voltage and frequency combination specified in 3.6 will not damage the spectrometer.

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3.6.1.2 Automatic standby switch. If a warmup or stabilization period is required, the instrument shall have an automatic standby device, that must be effective for at least 4 hours, so that in the event of a power outage, normal operation is possible when power is restored without a warmup or stabilization period.

3.6.1.3 If voltage variations of ±20 percent could cause damage to the spectrometer, safety devices shall be provided to automatically switch the spectrometer to the off mode at ±20 percent of nominal voltage.

3.6.2 Grounding. All electrical equipment shall be grounded in accordance with ANSI Standard C1-1968.

3.6.3 Motors. All motors shall conform to CC-M-636 or CC-M-641, as applicable. They shall be of adequate rating to operate under all conditions specified herein at maximum capacity without exceeding the NEMA recommended temperature limits for the class of insulating material used. The manufacturer's recommended full-load current shall not be exceeded during operation. Motors which may be exposed to an explosive-type atmosphere shall be of the explosion-proof type.

3.6.4 Wiring. All wiring between electrical components shall be in accordance with ANSI Standard C1-1968 and MIL-W-5088.

3.6.5 Electrical system components. The electrical system shall be equipped with the necessary inverters, rectifiers, transformers, et cetera, of sufficient power to operate all components normally operated simultaneously in the spectrometer. Solid-state components shall be utilized throughout the design of the spectrometer, except for the photoelectric tubes.

3.6.5.1 Electrodes. If the spectrometer uses rod-shaped electrodes, it shall accommodate electrodes of at least 6 inches in length.

3.7 Control panel. A control panel shall be provided and shall contain the instruments and controls necessary for convenient operation by the operator. A minimum 12- by 24-inch working area for recording data and pouring samples shall be attached to the front of the spectrometer convenient to the operator.

3.8 Control panel instruments

3.8.1 Instrument removal. The instruments shall be so designed and installed that each can be readily removed for maintenance and calibration. The instruments shall be removable from the front of the control panel, or by removal of a subpanel to gain access to the instrument mounting fixture.

3.8.2 Readability. Unless otherwise specified herein, all instrument scales shall be so selected that the smallest division spacings are not less than

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1/16 inch apart, and shall represent not more than one-half the specified accuracy limit (see 6.3.7). All essential dial faces on the control panel and readout windows shall be clearly visible to the operator from his normal operating position.

3.8.3 Instrument accuracy and calibration. The accuracy (see 6.3.7) of each instrument system shall be as specified herein. Unless otherwise specified herein, the system accuracy includes the accuracy of the complete system (ie, indicator, cable or connector and transmitter, transducer, or thermocouple). Zero adjustment features shall be provided on all instruments requiring calibration. Except for the standardization adjustment for the contaminant value indicator, all instrumentation shall have a minimum calibration period of 30 days.

3.8.4 Spectrometer calibration

3.8.4.1 A maximum of 15 minutes during an 8-hour period shall be permitted to determine whether the spectrometer is still within calibration.

3.8.4.2 Spectrometer calibration shall be accomplished by variable adjustment devices in lieu of adding or subtracting fixed value components. The calibration control and inputs shall be located on the spectrometer face. Zero and gain controls shall be provided to correct for parallel shift of all element curves.

3.9 System performance

3.9.1 Allowable deviation. The maximum allowable deviation for each element shall be:

$$\text{Dmax } (Y \leq 100) = \sqrt{b^2 + (c\bar{X})^2}$$

$$\text{Dmax } (Y > 100) = f\bar{X}$$

and the actual deviation of N (see 6.2) consecutive analyses of the same standard shall be calculated as follows:

$$\text{Dact} = \left[\frac{N\left(\Sigma Xi^2\right) - \left(\Sigma Xi\right)^2}{N\,(N-1)} \right]^{1/2}$$

Where

Dmax	Maximum allowable deviation
Dact	Actual deviation
Y	Nominal value of the standard in ppm
N	Number analyses per sample
Xi	The determined contamination value of each analysis
X	The average determined contamination value of N consecutive analyses
b,c,f	Element constants (see 3.9.1.1.1).

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3.9.1.1 Accuracy. The determination of contamination value of each element shall be made with the accuracy as expressed in the following equation:

$$Y - \bar{X} \quad max = \sqrt{.25 \qquad (.08Y)^2} + \Delta Y$$

Where Y = The nominal value of the standard in ppm
X = The average determined contamination value of N consecutive analyses
N = Number of analyses per sample (see 6.2)
ΔY = Accuracy tolerance of the standard oil in ±ppm (see 6.2).

3.9.1.1.1 Element constants. The constants in the equation in 3.9.1 are as follows:

Elements	b	c	f
Al, Be, Cr, Ni, Si	0.5	.05	.05
Fe	0.5	.05	.08
Cu, Mg	0.5	.05	.09
B, Ba, Cd, Mn, Ti	0.5	.06	.06
Ag, Mo	0.5	.08	.08
Na	0.5	.08	.16
Zn	0.5	.12	.12
V	0.8	.05	.05
Pb	0.9	.06	.06
Sn	1.0	.06	.06

3.9.1.1.1.1 Data reduction. For all repeatability calculations on Zn and for the repeatability calculations on the other elements above Y = 100, a maximum of three analytical results may be dropped, one at a time. A new deviation (D), average ($\bar{X}$), and max deviation (Dmax) shall be calculated following each drop. Thereafter, any analytical result that differs from $\bar{X}$ by ±1.91D where the number of data points remaining is 27 and ±1.84D where the number of data points remaining is 12 also may be dropped.

3.9.2 Indicator. One indicator shall be provided to indicate the contaminant value of each contaminant element within the sample. The indicator shall be an in-line electronic numerical tube readout type. The indicator shall be capable of indicating from 0 to 997 ppm for each of the 20 elements as desired. The instrument shall read 998 ppm whenever concentration of an element is higher than 997 ppm. The readout of any element shall not be affected by the presence or absence of any other element or combination of other elements.

3.9.2.1 All variable controls used for adjusting or calibrating the readout shall be mounted on the outside of the spectrometer.

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3.9.2.1.1 Range selector switch. When a two-range indicator is utilized, a two-position selector switch shall be provided for selecting the desired indicator range. The switch shall be mounted on the control panel or indicator.

3.9.2.2 Contaminant element selection. A selector shall be provided for selecting each desired contaminant to be displayed on the required indicator.

3.9.2.3 Standardization adjustment. A standardization adjustment control shall be provided for each element on the spectrometer. Each standardization adjustment control shall be mounted on the control panel in a manner that will prevent inadvertent movement and adjustment.

3.9.2.3.1 Internal reference channel. If an internal reference channel is used, it shall be set to the 4,861.3 - angstrom band structure occurring in the spectrum of hydrocarbon oils.

3.9.2.3.2 Reference standards. Standards for calibration and acceptance testing shall be those approved by the Tri-Service Calibration Standards Committee.

3.9.2.4 Alternate readouts. The spectrometer shall be so designed that it can be easily converted to drive associated printout equipment, ie, a slave typewriter or a digital computer as required for automatic data processing. If the output signal is used to drive a slave typewriter, the typewriter shall automatically print data onto a standard engine data card. The output receptacle shall be readily accessible from outside the spectrometer.

3.9.3 Associated printout equipment. Associated printout equipment shall be as specified by the procuring activity (see 6.2). When such printout equipment is required, all accessories required for mutually compatible operation shall be provided by the spectrometer supplier.

3.9.4 Elapsed time meter and analysing counter. An elapsed time meter capable of recording 9,999 operating hours minimum shall be mounted on the control panel. The meter shall meet all operational and nonoperational requirements specified herein. The elapsed time meter shall maintain the correct operating time while operating from any of the power sources specified in 3.6, but shall not include standby time at reduced power. Additionally, a seven-digit analysis counter shall be provided.

3.9.5 Noise levels. The noise levels produced by the spectrometer shall not exceed the sound pressure level values specified in table I when measured as specified in 4.6.13.

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TABLE I. Noise Level Requirements

Octave Band Center Frequencies (Hz)	Sound Pressure Levels (Decibels ref. 0.0002 dynes/cm^2)	
	15 Foot Radius	Operator's Position
63	71	78
125	65	72
250	59	67
500	54	64
1,000	52	62
2,000	50	60
4,000	48	58
8,000	47	57

3.10 Main power input. The main power input shall be made through an AN or MS connector mounted on the side of the spectrometer.

3.10.1 Main power input cable. A main power input cable shall be provided and shall be a minimum length of 20 feet. Connector plugs shall not be provided on the power cable free end(s) which mate with the commercial power source(s). The cable free end(s) shall be protected from fraying and contamination during shipment and while not in use.

3.11 Interconnecting cable and wire assemblies

3.11.1 Molded backshells. The main power input cable and cable connectors requiring this type backshell within the spectrometer shall be provided with backshells molded with compound conforming to MIL-S-8516. The mold shall seal the back of the connector and bond to the jacket, eliminate the necessity for a clamp, and strengthen the junction between the cable and connector.

3.11.2 Connectors. Where not otherwise specified, connectors conforming to MIL-C-5015 shall be used throughout the spectrometer. All connectors exposed to the elements shall be provided with dust and moisture caps.

3.12 Transit case. A transit case where specified shall be provided for shipment, storage, and unlimited use (see 6.2). The transit case shall be in accordance with MIL-T-21200, paragraph entitled Transit case, except as modified herein.

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3.12.1 Handles. Handles shall be provided on the transit case to remove the cover from the transit case and for moving the transit case less the spectrometer.

3.12.2 Lifting and tiedown fittings. Lifting and tiedown fittings shall be provided on the transit case to permit hoisting and tiedown of the spectrometer for safe handling and shipment using all modes of transportation.

3.12.3 Forklift openings. The base of the transit case shall provide forklift openings on all sides located symmetrically about the horizontal center of balance to permit insertion of forklift tines from all sides. The openings shall be a minimum of 2-1/2 inches high and 8 inches wide. The openings shall be spaced not less than 24 inches nor more than 30 inches center-to-center.

3.12.4 Relative humidity indicator. The transit case shall be provided with a humidity indicator plug in accordance with MIL-I-26860. The plug shall be installed on one end of the case, preferably adjacent to the pressure relief valve. Location of the humidity indicator plug shall not interfere with any fastening or handling device.

3.13 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.14 Electromagnetic interference. The spectrometer shall comply with the provisions and limits of MIL-STD-461 for class IIA equipment.

3.15 Threads

3.15.1 Screw threads. Screw threads shall conform to Handbook H28.

3.15.2 Pipe threads. Pipe threads shall conform to MIL-P-7105.

3.16 Dimensions. The spectrometer dimensions shall not exceed 60 inches in height, 60 inches in width, and 36 inches in depth.

3.17 Gross weight. The combined maximum dry gross weight of the spectrometer and transit case shall not exceed 1,500 pounds.

3.18 Finishes and protective coatings. Cleaning, painting, plating, and anodic films shall be in accordance with MIL-STD-808. The color shall be light blue, color 25414 of FED-STD-595.

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3.19 Operational markings

3.19.1 Transit case marking. The following shall be marked on the transit case near the handles:

FOR LIFTING TRANSIT CASE ONLY.

3.19.2 Nameplates, identification plates, and placards. All nameplates, identification plates, and placards shall be in accordance with MIL-P-514. All plates shall be attached by metal screws or rivets.

3.19.2.1 Vendor's nameplate. The vendor's standard nameplate may be installed in addition to the nameplate of the procuring activity.

3.19.2.2 Placards. All valves, instruments, controls, outlets, and connections shall be identified by a name-and-number placard. The same numbers shall be shown on the flow and circuit diagrams and other data furnished with the spectrometer.

3.19.2.2.1 Warning. Warning placards or other precautionary labels as necessary shall be so designed and located that the operator cannot fail to notice them before controls, valves, and switches are operated. Warning placards shall be of laminated, phenolic sheet with red-surface backing to result in a red placard with white, engraved letters.

3.19.3 Cable conductor identification. Conductors shall be numbered in accordance with MIL-W-5088. In the event of shielded or jacketed conductors, the individual conductors shall be color coded and the outer jacket numbered.

3.19.3.1 Cable and hose identification. Cable and hose assemblies shall be identified with nondestructible, nonremovable, nonmetallic tags inscribed with the cable and hose number mounted in place at each end of the hose and cable not less than 3 inches from the connector.

3.19.4 Stenciling

3.19.4.1 The following information shall be stenciled in 1-inch black letters on each of the four sides of the transit case:

FOR AIR SHIPMENT
THIS UNIT IS DESIGNED FOR

__*__ G FORE AND AFT

__*__ G SIDEWARD

* To be filled in by the contractor.

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3.19.4.2 The following information shall be stenciled in 1-inch black letters on each tiedown and lifting point:

TIEDOWN POINT
OR
LIFTING POINT

GROSS WEIGHT ___*___

* To be filled in by the contractor.

3.20 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.21 Workmanship

3.21.1 Riveting. Riveting operations shall be performed to insure tightness and satisfactory heading of rivets.

3.21.2 Cleaning. The spectrometer shall be thoroughly cleaned. Loose, spattered, or excess solder; metal chips; and other foreign materials shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the spectrometer shall be classified as follows:

a. Preproduction inspection . . . . . See 4.4

b. Quality conformance inspection . . . . . See 4.5.

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, all tests required by this specification shall be performed at an atmospheric pressure of 29.5 to 30 inches Hg, temperature of 70° to 75°F, and a relative humidity of 60 to 70 percent.

4.3.2 Preliminary run-in and calibration. Preliminary run-in of the spectrometer as may be recommended by the manufacturer shall be performed by the manufacturer prior to submission of the spectrometer for testing. A certified log of all preliminary running time and calibration data shall be prepared.

4.3.3 Type test. A type I test or a type II test shall be conducted prior to, during, or after the preproduction tests and quality conformance inspections as specified in table II. However, if a type II test is successfully completed without moving, adjusting, or repairing the spectrometer, the data obtained therefrom may be used as the data for the next type II test prior to the required test.

4.3.3.1 Type I test. A type I test consists of the following:

a. Functional equipment check (see 4.6.2.4)

b. Equipment standardization check (see 4.6.2.3)

c. Full range calibration test (see 4.6.2.1).

4.3.3.2 Type II test. A type II test consists of the following:

a. Functional equipment check (see 4.6.2.4)

b. Equipment standardization check (see 4.6.2.3)

c. Limited range calibration test (see 4.6.2.2).

4.3.4 Classes of demonstrations. The demonstration tests are divided into three classes as follows:

a. Class 1 - Demonstration of the ability of the spectrometer to provide minimum allowable performance while being subjected to test conditions.

b. Class 2 - Demonstration of the ability of the spectrometer to provide minimum allowable performance after being subjected to test conditions.

c. Class 3 - Functional demonstration to validate design characteristics.

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TABLE II. Class, Number, and Type Test

Test No.	Test Title	Test Paragraph	Class	Test Type II Before	No. and Type Test During	Test Type II After	Comments
A-1	Examination of product	4.6.1	3	No	None	No	
A-2	Serviceability	4.6.1.1	3	No	None	No	
A-3a	Calibration	4.6.2.1	1	No	1 type I	No	Test at standard temperature
A-3b	Calibration	4.6.2.1	1	No	3 type I	No	One test each at 50, 90, & 110°F
A-4	Low temperature	4.6.3.1	2	Yes	None	Yes	
A-5	High temperature	4.6.3.2	2	Yes	None	Yes	
A-6	Low pressure	4.6.3.3	2	Yes	1 type II	Yes	
A-7	Humidity	4.6.3.4	2	Yes	None	Yes	
A-8	Salt fog	4.6.3.5	2	Yes	None	Yes (2)	Test immediately after salt fog and also after 48-hour drying period
A-9	Dust	4.6.3.6	2	Yes	None	Yes	
A-10	Fungus	4.6.3.7	2	No	None	No	
A-11	Vibration (transportation)	4.6.3.8	2	Yes	None	Yes	
A-12	Shock (transportation)	4.6.3.9	2	Yes	None	Yes	
A-13	Explosive atmosphere	4.6.3.10	3	No	None	No	Test sample excitation chamber Review rest of spectrometer
A-14	Shipboard installed Roll and pitch	4.6.3.11(a)	1	Yes	1 type II	Yes	
A-15	Shipboard vibration	4.6.3.11(b)	1	Yes	3 type II	Yes	One test in each axial direction
A-16	Severe shipboard roll and pitch	4.6.3.11.1	2	Yes	None	Yes	
A-17	Shipboard shock	4.6.3.11.2	3	No	None	No	
A-18	Electromagnetic interference	4.6.4	3	No	None	No	
A-19	Tilted position	4.6.5	1	No	8 type II	No	One test at each position

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TABLE II. Class, Number, and Type Test (Cont'd)

Test No.	Test Title	Test Paragraph	Class	Test Type II Before	No. and Type Test During	Test Type II After	Comments
A-20a	Mobility (on roads)	4.6.6(a)	2	Yes	None	Yes	
A-20b	Mobility (off-roads)	4.6.6(b)	2	Yes	None	Yes	
A-21	Reliability	4.6.7 thru 4.6.7.1.1	3	No	1 type II per 24 hrs	No	Utilize two pre-production spectrometers
A-22	Safety	4.6.7.3	3	No	None	No	
A-23	Maintainability	4.6.8 thru 4.6.8.2	3	No	None	No	Spectrometer verification after maintenance will be by visual observation, functional equipment check, and equipment standardization
A-24	Installation	4.6.9	3	No	None	No	
A-25	Tiedown & lifting	4.6.10	3	No	None	No	
A-26	Different power	4.6.11	1	No	2 type II	No	
A-27	Voltage safety	4.6.12	3	No	None	No	
A-28	Acoustical noise	4.6.13 thru 4.6.13.2	3	No	None	No	
A-29	Associated printout equipment test	4.6.15	3	No	None	No	
B-1	Examination of product	4.6.1	3	No	None	No	
B-2	Calibration	4.6.2.1	1	No	1 type I per test	No	Test at standard temperature
B-3	Reliability	4.6.7.2	3	No	1 type I per 24 hrs	No	Number of items as specified in 4.5.2
B-4	Installation & checkout	4.6.14	1	No	1 type I followed by 3 type II	No	Testing to be conducted as specified by the procuring activity (see 6.2)

Notes:

1. Test numbers preceded by the letter "A" are preproduction tests.
2. Test numbers preceded by the letter "B" are tests on production items.

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4.4 Preproduction inspection (see 6.2)

4.4.1 Test sample. Three spectrometers and one transit case shall be subjected to the tests specified in 4.4.3 as applicable.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the following:

a. Individual tests

b. Sampling plan and test.

4.5.1 Individual tests. Each spectrometer and transit case shall be subjected to the following tests as applicable:

a. Examination of product

b. Calibration tests.

4.5.2 Sampling plan and tests

4.5.2.1 Periodic sampling. A minimum of two spectrometers shall be selected at random from every 25 or fraction thereof produced for production and subjected to the tests specified in 4.6.7.2. The preproduction spectrometers shall not apply to satisfy this requirement.

4.5.2.1.1 Rejection and retest. When a spectrometer selected from a production run fails to meet the specification, items on hand or produced later shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of this specification.

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4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Inspection methods

4.6.1 Examination of product. The spectrometer shall be inspected to determine compliance with the requirements specified herein with respect to parts, tightness of fittings, screws, bolts, workmanship, finish, materials, marking, dimensions, wire coding and neatness of wiring, identification of cables, and compliance with codes and certifications as described in 6.4.

4.6.1.1 Serviceability. The spectrometer shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, operation, and safety. Particular note shall be made with regard to provisions to prevent accumulation of dirt and moisture which may hinder servicing and operation.

4.6.2 Calibration tests

4.6.2.1 Full range. The spectrometer shall be tested for calibration to determine compliance with the performance requirements specified herein. Prior to commencing this test, the spectrometer shall be allowed to warmup. This test shall consist of N (see 6.2) analyses each of the reference standards. The standards shall be provided by the procuring activity (1 pint of each standard per each spectrometer).

4.6.2.1.1 Allowable deviation. The allowable deviation shall be calculated for each contaminant in each sample test run and for all temperature test runs. The allowable deviation shall be calculated as specified in 3.9.1.

4.6.2.1.2 Average contaminant level. The average contaminant level of the N (see 6.2) consecutive analyses at any concentration level shall be as calculated in 3.9.1.1.

4.6.2.2 Limited range. This calibration test shall consist of N (see 6.2) analyses of one of the 10, 30, 100, or 300 ppm standards, and shall be rotated during testing to give complete coverage of all the above standards during testing. The allowable deviation and average contaminant level shall be as specified in 4.6.2.1.1 and 4.6.2.1.2 respectively.

4.6.2.3 Equipment standardization check. The equipment standardization check shall determine whether the analytical transfer function is preserved and whether adjustments are necessary for small variations. This shall be accomplished by analyzing blank oil (for zero adjustment) and 100 ppm reference standard (for initial slope adjustment).

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4.6.2.4 Functional equipment check. The performance of major instrument functions shall be checked by the spectrometer self-check provisions without the need to perform an oil analysis. The performance checking shall be accomplished by simulating signal inputs at various points in the spectrometer and by using the spectrometer readout system to display the results of inserting the stimuli. No additional equipment shall be required for a functional equipment check.

4.6.3 Environmental tests. Unless otherwise specified, the spectrometer shall be subjected to the following tests performed in accordance with the specified procedures of MIL-STD-810. Upon completion of each test, the spectrometer shall be subjected to the applicable test as specified in table II. Failure of the spectrometer to operate and meet the accuracies specified during or after each exposure, failure of any item of assembly due to deterioration from exposures in these tests, or formation of any undesirable condition that cannot be remedied by routine maintenance shall be cause for rejection.

4.6.3.1 Low temperature. The spectrometer shall be subjected to low temperature storage in accordance with method 502, procedure I. The temperature shall then be raised to 72°F and stabilized. Power shall be applied to the spectrometer, and a type II test shall be performed. The spectrometer shall be placed in the standby operating mode or power off to the spectrometer while the temperature is being stabilized at a new temperature.

4.6.3.2 High temperature. The spectrometer shall be subjected to high temperature in accordance with method 501, procedure I. The temperature shall then be lowered to 72°F and stabilized. Power shall be applied to the spectrometer and a type II test shall be performed. The spectrometer shall be placed in the standby operating mode or power off to the spectrometer while the temperature is being stabilized at a new temperature.

4.6.3.3 Low pressure. The spectrometer shall be subjected to low pressure in accordance with method 500, procedure I.

4.6.3.4 Humidity. The spectrometer shall be subjected to humidity in accordance with method 507, procedure I.

4.6.3.5 Salt fog. The spectrometer shall be subjected to salt fog in accordance with method 509, procedure I for 50 hours. The spectrometer shall be so protected that salt fog will not spray directly onto it, but will permit full exposure to the salt fog atmosphere.

4.6.3.6 Dust. The spectrometer shall be subjected to dust in accordance with method 510, procedure I. The spectrometer shall be so protected that dust will not spray directly onto it, but will permit full exposure to the dust atmosphere.

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4.6.3.7 Fungus. Selected parts from the spectrometer shall be subjected to fungus in accordance with method 508, procedure I.

4.6.3.8 Vibration. While mounted in the transit case, the spectrometer shall be subjected to vibration, sinusoidal resonance, and cycling tests in accordance with method 514, procedure X, except the 1.3 acceleration level shall apply in the frequency range from 26 to 50 Hz.

4.6.3.9 Shock. While mounted in the transit case, the spectrometer shall be subjected to shock in accordance with method 516, procedure II.

4.6.3.10 Explosive atmosphere. The sample excitation chamber of the spectrometer shall be subjected to an explosive atmosphere in accordance with method 511, procedure I, except the test will be accomplished at ground level only. Additionally, a detailed safety review of the rest of the spectrometer shall demonstrate that there is no explosion hazard.

4.6.3.11 Shipboard. The spectrometer in the installed position shall be subjected to a type II test during the following conditions:

a. Simulated shipboard roll of 10° and 5° pitch, permanent list of 10° and 5° trim fore and aft

b. Vibration as specified in MIL-STD-167, type I.

4.6.3.11.1 The spectrometer in the installed position shall be subjected to a type II test after being subjected to simulated shipboard roll of 45°, pitch of 10°, and list of 10°.

4.6.3.11.2 The spectrometer in the installed position shall be subjected to the shock test as specified in MIL-S-901, grade B.

4.6.4 Electromagnetic interference. The spectrometer shall meet the electromagnetic interference requirements for class IIA of MIL-STD-461 when subjected to the tests as specified in MIL-STD-462.

4.6.5 Operation in tilted position. The spectrometer shall rest on a platform. The platform supporting the spectrometer shall be tilted in eight different directions 45° apart on a 5° incline with the true horizontal plane and a type II test performed at each direction.

4.6.6 Mobility. The spectrometer shall be installed in a government-furnished M119, 6-ton, two-wheel cargo semitrailer van and driven over roads as follows:

a. Unimproved roads at an average speed of 25 mph (maximum speed 30 mph) for a distance of 100 miles

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b. Off roads over rolling terrain at an average speed of 10 mph (maximum speed of 12.5 mph) for a distance of 100 miles

Upon completion of each test above, power shall be applied to the spectrometer and a type II test performed. Failure of the spectrometer to meet the requirements of a type II test shall be cause for rejection. The warmup time specified in 3.5 shall apply.

4.6.7 Reliability test and demonstration. Demonstration of compliance with the reliability requirements shall be in accordance with MIL-STD-781 with the exceptions and conditions specified below, including 4.6.7.1 and 4.6.7.2. The specified MTBF of 3.4.2 as defined in MIL-STD-781, is the value to be used to divide the accumulated operating time for accept-reject criteria.

a. The spectrometer shall be subjected to the calibration test (4.6.2.1) at each of the following temperatures: 50°, 72°, 90°, and 110°F. These temperature levels shall constitute the test level. Vibration or voltage cycling shall not apply during reliability testing. The deviation specified for repeatability and accuracy (4.6.2.1.1 and 4.6.2.1.2) shall be determined for each temperature, each element, and each of the samples and also for all of the test points for each element for each of the samples over the temperature range of 50° to 110°F (see 6.2)

b. Operating time or analyses accumulated during other tests specified herein may be counted toward reliability decision time. Added operating time or analyses needed shall be accumulated at conditions equivalent to the calibration tests at the temperatures indicated above. Satisfactory performance shall be verified not less than once in each 24-hour operating period by repeating and recording the results of the calibration test

c. The spectrometer shall be used at a rate of not less than 320 analyses per 8-hour period. The tests may be accelerated by operating at a faster use rate with the decision criteria established in analyses where 320 analyses are equivalent to 8 operating hours

d. Added test cycling shall be so established that operation at the high and low temperature extremes can be done with no intervening downtime. The spectrometer shall be placed in the standby operating mode while the chamber temperature is being stabilized at a new temperature, except that not less than once in each completion of test temperatures the spectrometer shall be shutdown and stabilized at room temperature before beginning the next temperature test

e. Longevity tests of MIL-STD-781 shall not apply

f. Malfunctions that may be considered minor in that they do not cause unacceptable performance degradation shall be included in the pattern failure corrective action provisions of MIL-STD-781.

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4.6.7.1 Preproduction reliability demonstration. Preproduction spectrometers, representative of expected production, shall be tested in accordance with 4.6.7 with the following conditions. These tests shall be considered the qualification tests of MIL-STD-781.

4.6.7.1.1 The tests and accept-reject criteria shall be in accordance with test plan XXVI (fixed length demonstration test) of MIL-STD-781. In determining the quantity required for test, integrated circuits shall count as the number of equivalent discrete parts in traditional wired circuitry that would replace the integrated circuits.

4.6.7.2 Sampling reliability demonstration. Production spectrometers shall be tested in accordance with 4.6.7 with the following conditions:

a. The tests and the accept-reject criteria shall be in accordance with test plan XXVII (fixed length production verification tests) of MIL-STD-781

b. For accept-reject decision purposes, data from as large a sample size as practicable within 45-day periods shall be grouped and applied (eg, two equipments of 25 produced if within a 45-day period)

c. In addition to the accept-reject criteria on a lot-by-lot basis as above, the data collected on samples taken during production shall be accumulated and an accept-reject decision made on the basis of accumulated data. The accept-reject criteria shall be in accordance with test plan XXVII of MIL-STD-781, except an accept decision is permissible even if pattern failures are indicated by the accumulated data of different sampling groups or lots. Where pattern failures are indicated by the accumulated data, the contractor shall propose appropriate corrective action to the procuring activity.

4.6.7.3 Safety demonstration. Compliance with the qualitative safety requirements shall be demonstrated by inspection, by the absence of hazards greater than class II of MIL-S-38130 during testing or inspection, and by a qualitative hazard analysis. The analysis may be part of or related to the reliability prediction. The level of the analysis need not go below the assembly or component level that reflects the potential hazard caused by any failure mode or by any operating mode of any part in the assembly or component.

4.6.8 Maintainability demonstration. Compliance with qualitative maintainability requirements shall be demonstrated by inspection of examination of product. Compliance with quantitative maintenance downtime requirements shall be demonstrated by the task performances specified below. The active maintenance downtimes, the tools and equipment required, and any obstructions or conditions hindering task performance or presenting potential hazard to personnel or to parts shall be observed and recorded. Tools and equipment

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recommended for maintenance shall be used to the maximum practicable extent during testing with emphasis on any special tools or equipment required. Use of special tools and equipment shall be sufficient to determine that they are useful and justified. Instances where special tools or equipment are inadequate shall be recorded in detail. Administrative and supply time, such as completing records or obtaining parts, need not count as active downtime. Maintenance task time data shall be collected wherever practicable from other tests or inspections specified herein and shall be applied to the maintainability demonstration wherever adequate observations have been made and recorded. The skill levels of maintenance technicians and operators performing maintenance during the demonstration shall approximate the contractor's recommendations of skill levels of military personnel to do similar tasks in service use of the equipment. Signs of excessive wear or other indications of potential failure shall be recorded where observed and the information supplied to the design activity for study and recommendation of appropriate modifications to be reflected in items to be delivered. Such recommendations shall be subject to procuring activity approval before being implemented or successful retest of other tests specified herein may be required where the test result could be influenced by the modifications. It shall be demonstrated that the equipment is in satisfactory operating condition after the maintenance tasks have been done. Maintenance tasks that may be practicable at depot level but that are not recommended by the contractor below depot level shall be identified and a prediction made of the downtime for each such task with the tools and equipment required. Such depot tasks need not be physically simulated. The predictions may be made based on engineering study and judgment, comparison to times for similar tasks, and study of equipment and drawings.

4.6.8.1 Corrective maintenance demonstration. Compliance with quantitative corrective maintenance downtime requirements shall be demonstrated for on-equipment maintenance by a maintenance demonstration procedure, maintenance task selection, and maintenance task performances in accordance with MIL-STD-471 including appendixes A and B, method 1, test plan pair A1 plus B1. Compliance with quantitative requirements for off-equipment maintenance (ie, components that are replaced in the spectrometer equipment by a substitute component then repaired while or after the equipment is returned to satisfactory condition) shall be demonstrated by predictions of maintenance times. The predictions shall note the assumptions made concerning the availability of spare parts and components.

4.6.8.2 Preventive maintenance demonstration. Compliance with preventive maintenance downtime requirements, both on-equipment and off-equipment, shall be demonstrated in accordance with MIL-STD-471, appendix B, method 6 by performing all such tasks. The tasks shall include all calibration tasks, such as instrument system calibrated, and all servicing tasks, such as fluid filters changed. Each task need not be done more than once. Task frequencies of occurrence shall be established before the demonstration tasks are done.

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4.6.9 Installation. It shall be demonstrated that the spectrometer can be removed from the transit case and installed for operation within 8 man-hours.

4.6.10 Tiedown and lifting. With the spectrometer mounted in the transit case, the transit case shall be subjected to the tests specified in MIL-A-8421. During this testing, the transit case shall be secured by means of the lifting and tiedown fittings only and each load shall be applied at least once. Upon completion of this test, the transit case shall be inspected to determine the adequacy of design and the security of mounting of all components.

4.6.11 Voltage and frequency range. The acceptance criteria for the requirements of 3.6 shall be the successful completion of type II performance demonstration test performed using 120V ac, 50 to 60 Hz; record voltages at the equipment test points at 120V ac, and compare these test point voltages at the other six input voltages.

4.6.12 Safety device. To demonstrate compliance with 3.6.1.1, the spectrometer shall be wired for operation from each of the specified voltages and frequencies and then the other specified voltages and frequencies shall be applied.

4.6.13 Noise level. All acoustical measurements shall be made during the burn cycle of the spectrometer. The noise levels produced by the spectrometer shall not exceed the sound pressure level values specified in 3.9.5 in any octave band at any of the measurement positions specified in 4.6.13.1. Ambient background noise levels at all measurement positions shall be at least 10 dB below the octave band sound pressure levels produced by the spectrometer. The acoustical measurement instrumentation shall be as specified in 4.6.13.2.

4.6.13.1 Measurement positions. Acoustical measurements shall be made at eight positions equally spaced every 45° on a 15-foot radius circle centered on the geometrical center of the spectrometer. Measurements also shall be made at three positions located 2 feet from the operator's control panel and spaced 2 feet apart. All measurements shall be made with the microphone 5 feet above the floor and oriented toward the spectrometer.

4.6.13.2 Instrumentation. The basic required instrumentation shall consist of a sound level meter in conjunction with an octave band analyzer. This instrumentation shall meet the requirements specified in ANSI standards S1.4 and S1.11, respectively. An up-to-date calibration and frequency response of the entire measurement system, including microphone, shall be available to permit correction of the measured data. The range of frequency response of the system shall be from 45 to 11,200 Hz and the response stated within $\pm$2 dB. An acoustical calibrator shall be used for an end-to-end system

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calibration from the microphone to the meter. The sound pressure level (in dB ref 0.0002 dynes/cm^2) produced by the calibrator, when used in conjunction with the particular microphone in the system, shall be known within $\pm$0.5 dB.

4.6.14 Installation and checkout. After all tests specified herein have been successfully performed, an installation and checkout test shall be performed by the contractor at the destination of the spectrometer as specified (see 6.2). This test will be installation of the spectrometer in not more than 8 man-hours (see 3.4.8) and performing one type I test at any temperature within the range of 50° to 110°F, followed by three type II tests. There shall be 1/2 hour of instrument standby between the last analysis of one test and the first analysis of the next test. The inability of the spectrometer to meet the installation requirements and to satisfactorily analyze element contaminants (see 3.9.1 and 3.9.1.1) in standard samples shall be cause for rejection.

4.6.15 Associated printout equipment. When specified (see 6.2), tests on associated printout equipment shall consist of a tape data system demonstration conducted at standard atmospheric conditions specified in 4.3.1. The demonstration shall consist of the complete analysis and data handling procedure for two oil samples (50 and 300 ppm standards). Data coding requirements and special data forms required for the demonstration shall be as provided by the procuring activity.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging for all levels shall be as specified in 5.1.1.

5.1.1 The spectrometer shall be preserved and packaged in accordance with method IId of MIL-P-116. The unit and shipping container shall be the transit case as specified in 3.12. The quantity of desiccant shall be determined in accordance with formula II of MIL-P-116 and securely placed in the transit case in a manner that will not cause damage to the unit.

5.2 Packing. Packing for all levels shall be as specified in 5.2.1.

5.2.1 The spectrometer, preserved and packaged in a transit case as specified in 5.1.1, shall not be overboxed. The transit case shall be used as the shipping container for all levels of packing required.

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5.3 Marking. In addition to any other marking required by the order or contract, the transit case shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The A/E35U-3 spectrometer is for use by organizational, direct support, general support, and depot maintenance personnel in determining wear, incipient failure, or the need for preventive maintenance of fluid wetted components by precise analysis of the operating fluid for wear metals, additives, and tracer elements.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification

b. What the accuracy tolerance (ΔY) is on the standards provided by the government (see 3.9.1.1)

c. Whether printout equipment is required and the type to be provided (see 3.9.3)

d. How many consecutive analyses (N) are performed (3.9.1, 3.9.1.1, 4.6.2.1, 4.6.2.1.2, and 4.6.2.2)

e. Quantity of transit cases required (see 3.12)

f. Location and conditions for preproduction inspection (see 4.4.1)

g. Whether the installation and checkout test is to be performed at the destination of the spectrometer (see 4.6.14)

h. What the tests shall consist of on the associated printout equipment (see 4.6.15)

6.3 Definitions

6.3.1 MTBF. The total operating time from the beginning of operation of the spectrometer(s) to completion of the test period divided by the total number of failures plus one additional failure if the test period does not end in a failure, or divided by the total number of failures if the test period ends in a failure.

6.3.2 Reliability failure. For the purpose of computing MTBF, a failure is defined as any malfunction which the operator cannot remedy using system

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controls normally accessible to him during routine operation, and which the operator cannot repair using on-equipment-material, tools, and parts within 5 minutes, and which causes:

a. Mission downtime for unscheduled maintenance

b. Damage to the system by continued use

c. A personnel safety hazard

d. Serious degradation of mission performance

Simultaneous related malfunctions are considered one failure. Malfunctions which do not affect mission performance (eg, a cracked glass, broken knob) will not be considered failures.

6.3.3 Achieved availability. Achieved availability is the probability that a spectrometer will operate at any given time and is computed as follows:

$$A_a = \frac{MTBM}{MTBM + \bar{M}} \times 100$$

where: MTBM is mean-time-between-maintenance and ready time during the same time interval
$\bar{M}$ is mean active maintenance downtime resulting from both preventive and corrective maintenance.

6.3.4 Analysis. One analysis is defined as analyzing one sample to determine the contaminate value (in parts per million) of each of the 20 elements specified herein.

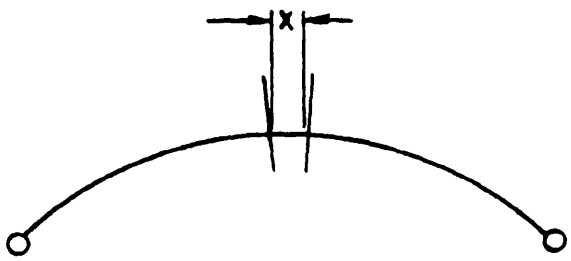
6.3.5 Downtime. That element of time during which the item is not in condition to perform its intended function.

6.3.6 Standby period. The standby period is the period where electrical power is supplied to the spectrometer to warmup the photoelectric tubes whose electric properties are modified by the action of light.

6.3.7 One-half the specified system accuracy. One-half the specified system accuracy means that if the specified system accuracy is $\pm$2 percent, one-half the specified system accuracy will be $\pm$4 percent.

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6.3.7.1 Example of readability. If a pressure system has a range of 10 to 100 psi with a system accuracy of 2 percent of full scale which equals 2 psi, scale divisions on the indicator should be spaced not less than 1/16 inch apart and represent not more than 4 psi as shown below.



X = ONE SCALE DIVISION
X ≥ 1/16 INCH
X ≤ 4 PSI

6.3.8 Accuracy. Unless otherwise specified, accuracy of an instrument system means the system accuracy of the instrument and transmitter combined and working as a single unit.

6.3.9 Calibration. Calibration is a series of operations or adjustments necessary to make instrument readout coincide with reference standard values.

6.3.10 Standardization. Standardization is the process of insuring that the spectrometer is in calibration.

6.3.11 Organizational maintenance. Organizational maintenance is that maintenance which is the responsibility of and performed by a using organization on its assigned equipment. Its phases normally consist of inspecting, servicing, lubricating, adjusting, and the replacement of parts, minor assemblies, and subassemblies. The Department of Defense term Organizational Maintenance encompasses the military services' maintenance terms of upkeep, line, hangar, organizational, and second echelon.

6.3.12 Intermediate maintenance. Intermediate maintenance is that maintenance which is the responsibility of and performed by designated maintenance activities for direct support of using organizations. Its phases normally consist of calibration, repair or replacement of damaged or unserviceable parts, components or assemblies; the emergency manufacture of nonavailable parts; and providing technical assistance to using organizations. Intermediate maintenance is

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normally accomplished in fixed or mobile shops, tenders, or shore based repair facilities. The Department of Defense term Intermediate Maintenance encompasses the military services' maintenance terms of field, minor modification, upkeep, voyage repairs, restricted availability, shops C and D, and third and fourth echelon.

6.3.13 Depot maintenance. Depot maintenance is that maintenance which is the responsibility of and performed by designated maintenance activities, to augment stocks of serviceable materiel, and to support organizational maintenance and intermediate maintenance activities by the use of more extensive shop facilities, equipment, and personnel of higher technical skill than are available at the lower levels of maintenance. Its phases normally consist of repair, modification, alteration, modernization, overhaul, reclamation, or rebuild of parts, assemblies, subassemblies, components, and end items; the emergency manufacture of nonavailable parts; and providing technical assistance to using activities and intermediate maintenance organizations. Depot maintenance is normally accomplished in fixed shops, shipyards, and shore based facilities. The Department of Defense term Depot Maintenance encompasses the military services' maintenance terms of rehabilitation, depot, fifth echelon, O&R, regular overhaul, restricted availability, and shops A and B.

6.4 Certifications

6.4.1 Electrical certifications. Proof should be submitted that the electrical system (3.6) and all motors specified which will be supplied under this specification meet the requirements of ANSI standard C1-1968 and the standards of Underwriters' Laboratories, Inc. with regard to fire and casualty hazards. The label or listing of the Underwriters' Laboratories, Inc., will be accepted as evidence of conforming to the specification requirements.

6.4.1.1 In lieu of the label or listing, the contractor may submit independent proof, satisfactory to the procuring activity, that the equipment conforms to the requirements of ANSI standard C1-1968 and the standards of Underwriters' Laboratories, Inc. with regard to fire and casualty hazards. Compliance with these requirements does not absolve the contractor from complete compliance with the other requirements of this specification.

6.5 Specification applicability date. The date of applicability of the documents listed herein is June 1969.

6.6 This revision covers the requirements of the A/E35U-3 spectrometer procured on contract N00600-70-C-1090.

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6.7 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Reviewers:
Army - MU
Navy - SH
Air Force - 82

Preparing activity:
Air Force - 11

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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